

AMD DEVELOPER HACKATHON: ACT II · TRACK 2

TextSink

Four voices. One truth.

FORMAL

SARCASTIC

HUMOROUS-TECH

HUMOROUS-NON-TECH

Built end-to-end on Gemma 4 · Fireworks AI

Two jobs. Most entries do one.

The LLM judge scores every caption on ACCURACY and TONE — across four styles.

Accurate but flat

Frame-only bots describe the scene correctly — then miss the joke. Tone score dies.

Funny but wrong

Prompt-only bots write a punchy line — about objects and events never on screen. Accuracy dies.

TextSink is built to win **both axes at once.**

One clip in. Four voices out.

Official judge clip — all four captions written live by Gemma 4.



FORMAL

broadcast precision

SARCASTIC

deadpan irony

HUMOROUS-TECH

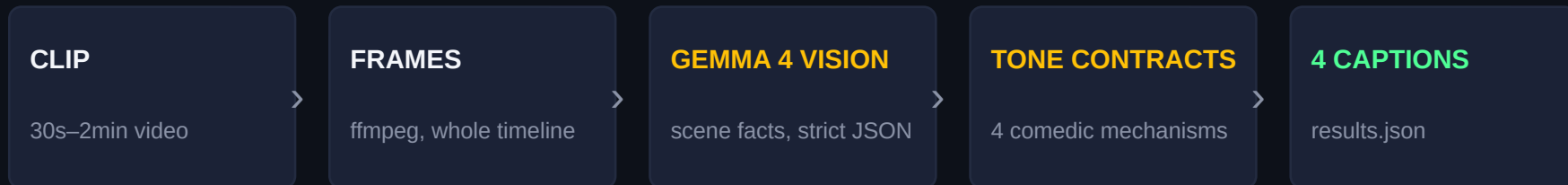
dev-culture metaphor

HUMOROUS-NON-TECH

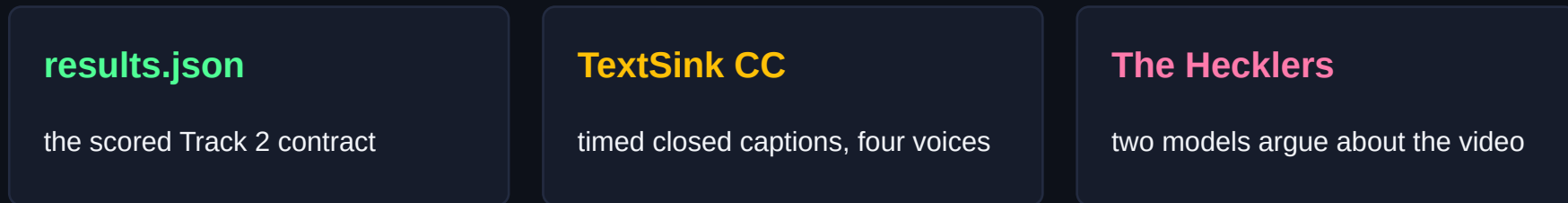
warm relatability

How it works

Ground first. Then let the voices loose.



The same grounding powers three products:



Accuracy is a discipline, not a hope.

Captions may only use extracted facts — and an empty output is structurally impossible.

Ground	Gemma 4 extracts scene facts as strict JSON from frames sampled across the whole clip.
Refuse	Empty grounding? Retry at rising temperature — then refuse to caption blind. Never hallucinate.
Survive	Three-tier ladder: Gemma deployment → serverless models → direct frame captioning. No task scores zero.
Parse anything	Schema drift, tool-call wrappers, markdown lists, compressed timestamps — all normalized.

Verified in-container: **3 official clips → 12 captions in 22 seconds** — far inside the 10-minute harness budget.

Four contracts, four voices.

Same grounded scene — the official kitten clip. Real, verified Gemma 4 output.

FORMAL

“An orange tabby kitten walks forward through the foliage toward the camera in a wooded area.”

SARCASTIC

“A fierce predator emerges from the brush, clearly ready to conquer the entire forest one tiny, uncoordinated step at a time.”

HUMOROUS-TECH

“The latest AI agent successfully navigating its first training environment. It’s small, but the feature set is looking promising.”

HUMOROUS-NON-TECH

“A tiny orange explorer makes a grand entrance, bravely navigating the treacherous jungle of backyard leaves.”

TextSink CC — real closed captions.

Timed .srt tracks that watch the whole video — one consistent voice per style, beat by beat.



The menace is gaining ground.

SARCASTIC TRACK

- 0:00 A tiny orange predator prepares for total world domination.
- 0:03 The terrifying advance begins through the dense shrubbery.
- 0:06 Truly, a relentless force of nature approaches.
- 0:09 The menace is closing in. Send help.

Every track spans the full clip — running gags included.

The Hecklers — two models, one argument.

Every line generated live, turn by turn: Gemma 4 vs gpt-oss-120b, each reading the other's last remark.



STAN: The deep blue menace is swallowing the very earth!

STAN: Ocean's attacking the coast! It's a watery apocalypse, Gus!

GUS: Apocalypse? Your kettle's just doing a light rinse.

STAN: The deep blue menace is swallowing the very earth!

GUS: Only you think foam is world-ending, Stan.

Three duos: **STAN & GUS** (sarcastic) · **LINT vs VIBE** (tech) · **DORIS & PEARL** (everyday)

Gemma 4, end to end — even the teacher.

One 26B MoE model on a dedicated Fireworks deployment does every creative job.

SEES	vision grounding of every clip
WRITES	all four caption styles + CC tracks
ARGUES	plays lead Heckler vs gpt-oss
JUDGES	scores its own drafts on accuracy + tone
TEACHES	judged winners became a self-distilled SFT dataset — Gemma teaches Gemma

\$5,500

addressable with one project

\$2,500 Track 2
\$3,000 Best Use of Gemma

TextSink

Four voices. One truth.

run the graded contract

```
docker run --rm \
  -e FIREWORKS_API_KEY=$KEY \
  -v $PWD/input:/input:ro \
  -v $PWD/out:/output \
  textsink
```

github.com/banksythequantLab/textsink

MIT · Python 3.11 · ffmpeg · Docker linux/amd64
Gemma 4 on Fireworks AI · AMD ACT II